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Ngu Dang

Summary

Recent Computer Science Ph.D. graduate specializing in **computational complexity**, particularly circuit complexity and its connections to **meta-complexity**. My research focuses on structural characterizations of optimal Boolean circuits, including fan-in/fan-out behavior, wiring patterns, and compositional forms, and explores how these characterizations can inform meta-complexity questions for explicit functions. In addition to my theory background, I have developed focused projects in machine learning, data science, and NLP, building on earlier undergraduate research in computer vision and demonstrating adaptability across new technical domains.

Education

Department of Computer Science, Boston University — Boston, MA
Ph.D. in Computer Science — 09/2020 - 05/2026

- Advisor: Prof. Steven Homer.
- Research area: Algorithms Design, Circuit Complexity, and The Minimum Circuit Size Problem (MCSP).
- Thesis: On Characterizing Optimal Boolean Circuits and Their Applications.
- GPA: 3.95/4.00.

Department of Computer Science, Clark University — Worcester, MA
B.A. in Computer Science, Minors: Data Science and Mathematics. — 01/2018 - 05/2020

- Advisor: Prof. Frederic Green.
- GPA: 3.93/4.00 — Graduated with Summa Cum Laude and High Honors.
- First Honors Dean's List in 2018, 2019, and 2020.

Professional Experience

Graduate Research Assistant | Boston University — 09/2020 - 05/2026

- Worked on joint projects on computational lower bounds and upper bounds of complex algorithms and designed improvements on top of state-of-the-art results.
- Implemented our algorithms, experiments, and scripts simulating Boolean circuits in Python using Z3 SAT-solver library.

Undergraduate Research Assistant | Clark University — 05/2019 - 05/2020

- Contributed to computer vision and computational geometry research projects for the Computer Science Department.
- Implemented experiments, statistical analysis (e.g. ANOVA, Kruskal-Wallis), visualization, and geometrical simulations in Python and Java.

Publications & Preprints

1. Marco Carmosino, **Ngu Dang**, and Tim Jackman. **Simple Circuit Extensions for XOR in PTIME.** *In 43rd International Symposium on Theoretical Aspects of Computer Science (STACS 2026). Leibniz International Proceedings in Informatics (LIPIcs), Volume 364, pp. 23:1-23:20, Schloss Dagstuhl – Leibniz-Zentrum für Informatik (2026).* DOI.
2. Marco Carmosino, **Ngu Dang**, Tim Jackman. **Constructive Separations from Gate Elimination**. 2026. Preprint. Under submission to MFCS 2026.
3. Marco Carmosino, **Ngu Dang**, Tim Jackman. **Convergent Gate Elimination and Constructive Circuit Lower Bounds**. 2026. Preprint.
4. Mariah Papy, Duncan Calder, **Ngu Dang**, Aidan McLaughlin, Breanna Desrochers, and John Magee. **Simulation of Motor Impairment with "Reversed Angle Mouse" in Head-Controlled Pointer Fitts's Law Task.** *In Proceedings of the 21st International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '19); ACM, Pittsburgh, PA, USA.* DOI.

Teaching Experience

Teaching Fellow | Boston University — 01/2021 - 12/2025

- CS131: Combinatorics Structures — Summer 2022, 2023.
- CS132: Geometric Algorithms — Summer 2022.
- CS235: Algebraic Algorithms — Spring 2021, Fall 2025
- CS237: Probability in Computing — Summer 2024.
- CS332: Theory of Computation — Spring 2023, Fall 2023, 2024.
- CS630: Advanced Algorithms — Fall 2021.

Grader | Boston University — 09/2023 - 01/2024

- CS535: Complexity Theory — Fall 2023.

Undergraduate Teaching Assistant | Clark University — 09/2018 - 12/2019

- CS120: Introduction to Computer Science — Fall 2018.
- CS121: Data Structures — Spring 2019.
- CS180: Automata Theory — Fall 2019.

Other Projects

Tweet Dialect Classifier

. *Personal Project* — *Github Link* — 06/2025 - 07/2025

- Built a dialect classifying pipeline in Python with BERTweet-based model that distinguishes African American Vernacular English from Standard and regular African American English and achieved 0.95, 0.99, and 0.97 for accuracy, recall, and F1 score respectively.
- Integrated the classifier into a bias-aware sentiment analysis pipeline, with statistical analysis (Kruskal-Wallis H Test) to provide insights on fairness in interpretation of social media text across different models (i.e. RoBERTa, RoBERTa-Latest, BERTweet).

Real-Time Object Detector

. *Personal Project* — *Github Link* — 05/2025 - 06/2025

- Built a real-time object detection system by training YOLOv8 on Pascal VOC (Python) and implementing C++ ONNX Runtime inference with OpenCV for webcam-based detection.
- Applied transfer learning with pretrained YOLOv8n weights and integrated ONNX Runtime C++ API to deliver fast, resource-efficient object detection with dynamic bounding box visualization and minimal latency.

Human Activity Recognition Using Deep Learning

. *Personal Project* — *Github Link* — 04/2025 - 05/2025

- Built a deep learning pipeline using Python and PyTorch to classify human activities from Wi-Fi CSI data, achieving 0.98 accuracy score with a custom CNN-LSTM model.
- Designed a complete preprocessing workflow including reshaping, normalization, smoothing, and statistical feature augmentation to improve model robustness.

Churn Predictor for Subscription Service

. *Coursera's Challenge* — *Github Link* — 03/2025 - 04/2025

- Implemented an end-to-end churn prediction pipeline in Python for a video streaming service using a real-world imbalanced subscription dataset using an ensemble of three models — a neural network, XGBoost, and Random Forest — using weighted soft voting to optimize class ranking and maximize AUC.
- Engineered advanced features (e.g., ratio metrics, interaction terms behavioral buckets, etc.) on top of 20 given features to boost signal quality and improve model discrimination and achieved achieved a ROC AUC score of 0.75.

Skills

Programming: Python, Java, C++, MySQL.

Libraries: Pandas, Numpy, Scipy, Tensorflow, PyTorch, Natural Language Toolkit (NLTK), Keras, Scikit-Learn, Seaborn, Z3.

Tools: LLMs, Git, Jupyter Notebook, Google Colab, Visual Studio, Microsoft Office Suite

Scripting: LaTeX, HTML, CSS.

OS: Windows, Linux.

Certificates

- **IBM Data Science by IBM on Coursera**. Certificate earned on 08/31/2023.
- **Neural Networks and Deep Learning by DeepLearning.AI on Coursera**. Certificate earned on 12/31/2024.

Awards and Honors

- **Outstanding Academic Achievements**, awarded by the Department of Computer Science at Clark University.
- **Inducted to Phi Beta Kappa** by Lambda of Massachusetts at Clark University on 05/24/2020.

Academic Services

Reviewer for: *Journal of Computer and System Science (JCSS)*.

Organizer for: *Boston University Computer Science's Theory Seminar (Spring 2021)*.

Vice President for: *Clark University Computer Science's Competitive Programming Club*.